

Mainul Islam

Researcher | Engineer | Robotics, Control & Machine Learning

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Research Interests

AI-enabled digital twins, control co-design (CCD), reinforcement learning for control, model predictive control, sensor fusion and state estimation, autonomous and robotic systems, and physics-based / data-driven modeling of dynamic systems.

Education

B.Sc. in Electrical and Electronic Engineering (EEE)

Jan 2020 – Jun 2024

Islamic University of Technology (IUT), Gazipur, Bangladesh

CGPA: 3.87 / 4.00

Thesis — MonoSIM: Developed an open-source Software-in-the-Loop (SIL) framework for autonomous Ackermann vehicles using a monocular-vision, sliding-window lane-detection pipeline, and implemented and benchmarked Model Predictive Control (MPC) and PID strategies for path tracking within the simulation environment (published, IEEE ICARA 2026). Supervisor: Dr. Golam Sarowar, Professor, Dept. of EEE, IUT.

Publications

[Google Scholar link](#)

Peer-Reviewed Journal Article

- A. U. R. Adib, **M. Islam**, M. S. Abid, R. Ahshan, “A deep learning based framework for solar panel segmentation and fault classification enhanced with explainable AI,” *Solar Energy*, vol. 302, 2025. (Impact Factor: 6.6)
DOI: [10.1016/j.solener.2025.114058](https://doi.org/10.1016/j.solener.2025.114058)

Peer-Reviewed Conference Papers

- S. Rahman, N. Hasin, **M. Islam**, M. Z. A. Rony, G. Sarowar, “MonoSIM: An open-source SIL framework for Ackermann vehicular systems with monocular vision,” *IEEE 12th Intl. Conf. on Automation, Robotics and Applications (ICARA)*, 2026. (**B.Sc. Thesis**)
DOI: [10.1109/ICARA69401.2026.11480327](https://doi.org/10.1109/ICARA69401.2026.11480327)
- **M. Islam** *et al.*, “A robust deep learning framework for Bangla license plate recognition using YOLO and vision-language OCR,” *IEEE Intl. Conf. on AI and Data Analytics (ICAD)*, 2026.
DOI: [arXiv:2603.10267](https://arxiv.org/abs/2603.10267)
- H. Haque, **M. Islam**, M. A. I. Alvy, M. S. Islam, M. A. Razzak, “Long-term energy demand analysis using machine learning algorithms: A case study in Bangladesh,” *IEEE ICECIE*, 2024.
DOI: [10.1109/ICECIE63774.2024.10815643](https://doi.org/10.1109/ICECIE63774.2024.10815643)
- **M. Islam** *et al.*, “Optimizing master-slave broadcast and unicast communication in multi-node STM32 networks using UART and RS485 protocols,” *IEEE ICCIT*, 2024.
DOI: [10.1109/ICCIT64611.2024.11022430](https://doi.org/10.1109/ICCIT64611.2024.11022430)

Teaching Experience

Part-Time Lecturer, Department of EEE — Ahsanullah University of Science and Technology (AUST)

Jul 2025 – Jun 2026

- Delivered undergraduate coursework and laboratory instruction; mentored students on projects spanning control, electronics, and applied programming, including course preparation, lecturing, and student assessment.

Professional Experience

Research and Development Engineer / Embedded Systems Engineer — Cybernetics Hi-Tech Solutions Pvt. Ltd. ([link](#)) Oct 2024 – Present

- **Autonomous Ground Vehicle (AGV):** Co-developed a from-scratch Extended Kalman Filter fusing dual-camera and wheel-odometry data with AprilTag-based visual pose estimation (roll/pitch/yaw); implemented LiDAR-based obstacle avoidance, Modbus encoder feedback, and coordinate-frame transformations linking robot, camera, and tag frames. Integrated path planning, localization, and motion control into a unified ROS2 autonomy stack. Diagnosed and resolved a high-traffic ROS2 network congestion issue caused by high-frequency, large-payload topics, and automated the stack for boot-time startup.
- **Man-Portable Teleoperated Robot:** Designed a fault-tolerant CAN-bus layer controlling 11 motors with heartbeat, fault, and battery-based autonomous response, and a ROS2 middleware layer above it that bridged two CAN networks, unified inverse-kinematics control and discrete command mapping. Built a fiber-optic/wireless failover communication link for SBUS control data. Developed a custom inverse-kinematics solver for a 4-DOF subsystem of a 7-DOF arm (turret, shoulder, wrist, telescopic joint); engineered a low-latency (<280 ms) 5-camera RTSP application with real-time 3D URDF/STL visualization, touchscreen gimbal control, dual-channel audio, and full motor/register diagnostics.
- **Toll-Booth Automation:** Developed event-driven barrier control (vehicle detection, payment, theft detection) with loop-coil vehicle sensing and license-plate recognition; designed and manufactured a compact custom PCB with tamper protection.

Technical Skills

Programming: Python, C/C++, MATLAB, Embedded C

ML / DL: PyTorch, TensorFlow, scikit-learn, YOLO, Explainable AI (LIME/SHAP), Vision-Language OCR

Robotics / Control: ROS2, Extended Kalman Filter (EKF), sensor fusion, inverse kinematics, MPC, PID, LiDAR/vision perception, AprilTag/vision-based localization, CAN bus, Modbus, SBUS

Modeling / Simulation: Software-in-the-loop (SIL) simulation, monocular vision, URDF/STL 3D visualization

Embedded & Communication Protocols: CAN Bus, RS-485, UART, Modbus, STM32, PCB design, embedded systems, real-time distributed control

CAD / 3D Design: Autodesk Fusion, Onshape

Tools: Git, Linux, RTSP/real-time video streaming, LaTeX

Honors & Achievements

- **IRC 2024 — 6th Global:** International Rover Challenge, ABEX Best Science Team (Project Altair, 1st in Bangladesh).
- **ERC 2023 — 16th Global:** European Rover Challenge (Project Altair).
- **IRDC 2021 — 11th Global:** Indian Rover Design Challenge (Team Avijatrik, 1st in Bangladesh).
- **MATLAB Cody:** Global rank 117; 914 problems solved; 26 badges. [[Profile](#)]

References

Prof. Dr. Golam Sarowar — *B.Sc. Thesis Supervisor*

Professor, Department of EEE, Islamic University of Technology (IUT)

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